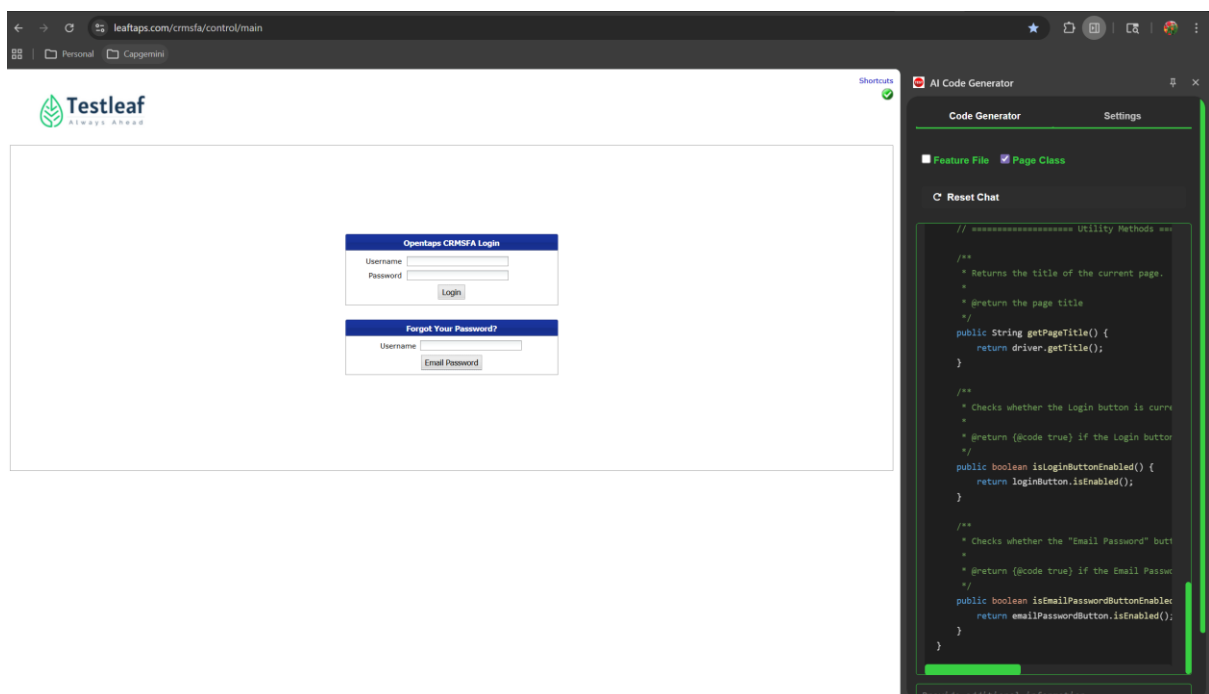
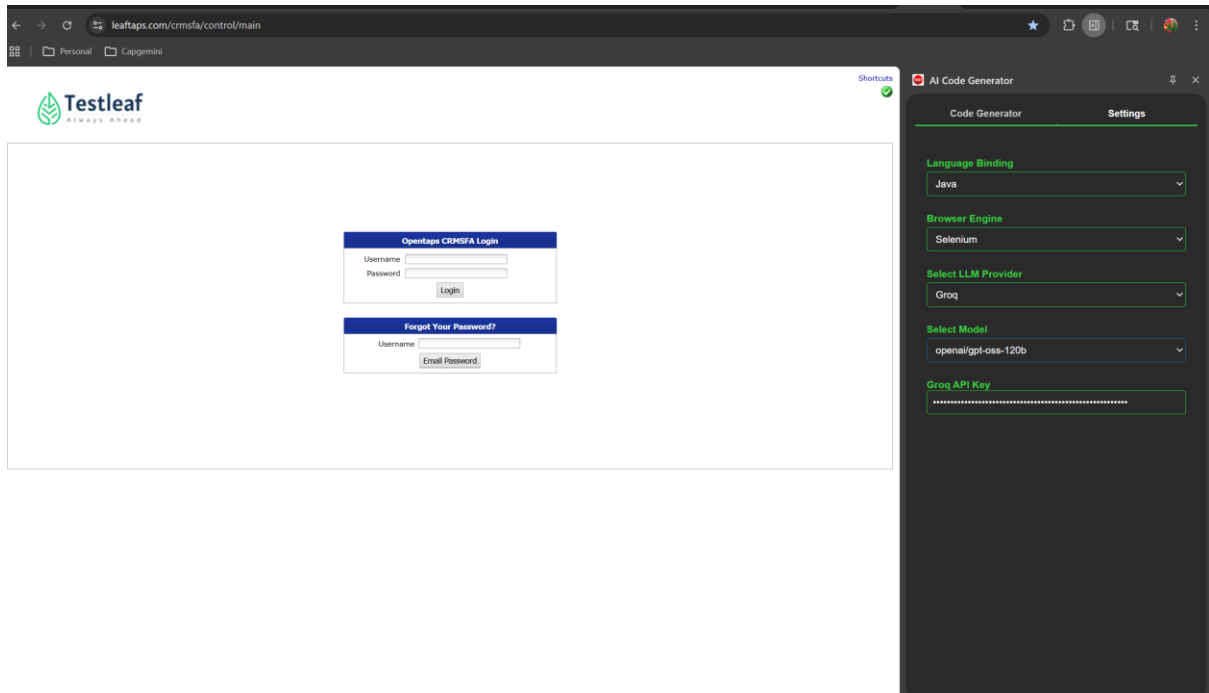


# Comparison of Code generated between standard model and fine-tuned model

## using

### AI Extension

#### 1. Code generated using standard model.



Generated code

```
package com.testleaf.pages;
```

```

import org.openqa.selenium.WebDriver;

import org.openqa.selenium.WebElement;

import org.openqa.selenium.support.FindBy;

import org.openqa.selenium.support.How;

import org.openqa.selenium.support.PageFactory;

/**
 * Page Object representing the Opentaps CRMSFA Login page.
 * <p>
 * Provides actions for the login form as well as the "Forgot Your Password?" section.
 * </p>
 */
public class OpentapsLoginPage {

    private final WebDriver driver;

    // ----- Login Form Elements -----

    @FindBy(how = How.XPATH, using =
"//form[@name='loginform']/input[@name='USERNAME']")
    private WebElement loginUsernameInput;

    @FindBy(how = How.XPATH, using =
"//form[@name='loginform']/input[@name='PASSWORD']")
    private WebElement loginPasswordInput;

    @FindBy(how = How.XPATH, using =
"//form[@name='loginform']/input[@type='submit' and @value='Login']")
    private WebElement loginButton;

```

```

// ----- Forgot Password Form Elements -----

@FindBy(how = How.XPATH, using =
"//form[@name='forgotpassword']/input[@name='USERNAME']")

private WebElement forgotUsernameInput;


@FindBy(how = How.XPATH, using =
"//form[@name='forgotpassword']/input[@type='submit' and @value='Email
Password']")

private WebElement emailPasswordButton;


/**
 * Constructor. Initializes web elements using {@link PageFactory}.
 *
 * @param driver the WebDriver instance controlling the browser
 */
public OpentapsLoginPage(WebDriver driver) {
    this.driver = driver;

    PageFactory.initElements(driver, this);
}


// ===== Login Form Actions =====


/**
 * Enters the supplied username into the login form.
 *
 * @param username the user name to enter
 */
public void enterLoginUsername(String username) {

```

```
loginUsernameInput.clear();  
loginUsernameInput.sendKeys(username);  
}
```

```
/**  
 * Enters the supplied password into the login form.  
 *  
 * @param password the password to enter  
 */
```

```
public void enterLoginPassword(String password) {  
    loginPasswordInput.clear();  
    loginPasswordInput.sendKeys(password);  
}
```

```
/**  
 * Clicks the Login button to submit the login form.  
 */  
public void clickLoginButton() {  
    loginButton.click();  
}
```

```
/**  
 * Performs a full login operation using the given credentials.  
 *  
 * @param username the user name  
 * @param password the password  
 */  
public void loginAs(String username, String password) {
```

```

    enterLoginUsername(username);

    enterLoginPassword(password);

    clickLoginButton();
}

// ===== Forgot Password Actions =====

/**
 * Enters the supplied username into the "Forgot Your Password?" form.
 *
 * @param username the user name whose password should be emailed
 */
public void enterForgotUsername(String username) {
    forgotUsernameInput.clear();

    forgotUsernameInput.sendKeys(username);
}

/**
 * Clicks the "Email Password" button to submit the forgot-password request.
 */
public void clickEmailPasswordButton() {
    emailPasswordButton.click();
}

/**
 * Requests a password reminder for the given username.
 *
 * @param username the user name to request a password for

```

```

*/

public void requestPasswordReminder(String username) {
    enterForgotUsername(username);
    clickEmailPasswordButton();
}

// ===== Utility Methods =====

/**
 * Returns the title of the current page.
 *
 * @return the page title
 */
public String getPageTitle() {
    return driver.getTitle();
}

/**
 * Checks whether the Login button is currently enabled.
 *
 * @return {@code true} if the Login button is enabled; {@code false} otherwise
 */
public boolean isLoginButtonEnabled() {
    return loginButton.isEnabled();
}

/**
 * Checks whether the "Email Password" button is currently enabled.

```

```

*

* @return {@code true} if the Email Password button is enabled; {@code false}
otherwise

*/

public boolean isEmailPasswordButtonEnabled() {

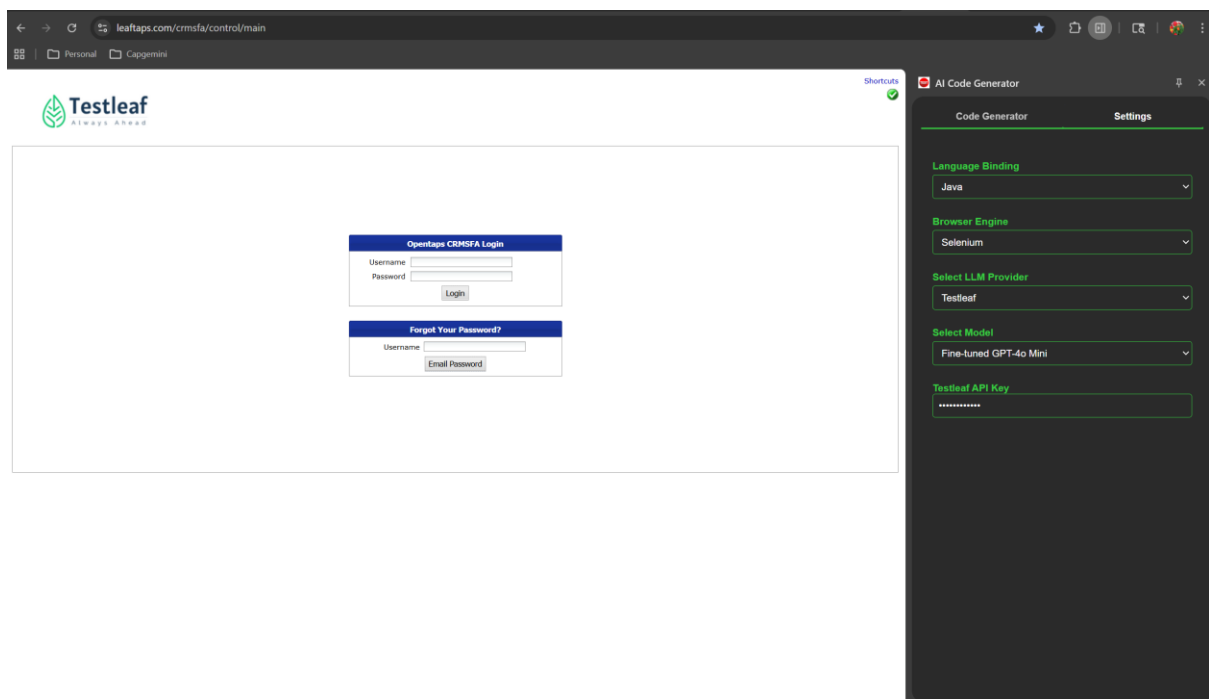
    return emailPasswordButton.isEnabled();

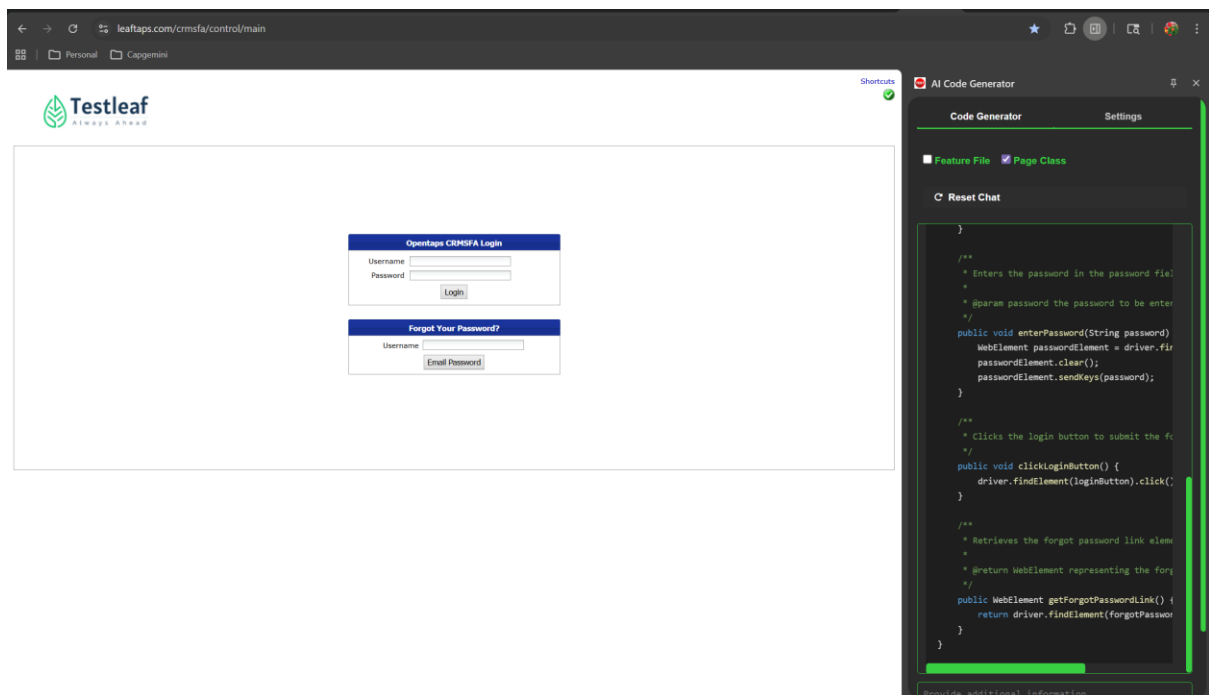
}

}

```

## 2. Code generation using Fine Tuned Model.





Generated code

```
package com.testleaf.pages;
```

```
import org.openqa.selenium.By;
```

```
import org.openqa.selenium.WebDriver;
```

```
import org.openqa.selenium.WebElement;
```

```
/**
```

```
 * Page Object for the Opentaps CRMSFA Login Page
```

```
 */
```

```
public class LoginPage {
```

```
    private WebDriver driver;
```

```
    // Locators
```

```
    private By usernameField = By.name("USERNAME");
```

```
    private By passwordField = By.name("PASSWORD");
```

```
    private By loginButton = By.cssSelector(".loginButton");
```



```
private By forgotPasswordLink = By.name("forgotpassword");
```

```
/**
```

```
 * Constructor to initialize the LoginPage with WebDriver
```

```
 *
```

```
 * @param driver WebDriver instance
```

```
 */
```

```
public LoginPage(WebDriver driver) {
```

```
    this.driver = driver;
```

```
}
```

```
/**
```

```
 * Enters the username in the username field
```

```
 *
```

```
 * @param username the username to be entered
```

```
 */
```

```
public void enterUsername(String username) {
```

```
    WebElement usernameElement = driver.findElement(usernameField);
```

```
    usernameElement.clear();
```

```
    usernameElement.sendKeys(username);
```

```
}
```

```
/**
```

```
 * Enters the password in the password field
```

```
 *
```

```
 * @param password the password to be entered
```

```
 */
```

```
public void enterPassword(String password) {
```

```
        WebElement passwordElement = driver.findElement(passwordField);

        passwordElement.clear();

        passwordElement.sendKeys(password);
    }

    /**
     * Clicks the login button to submit the form
     */
    public void clickLoginButton() {
        driver.findElement(loginButton).click();
    }

    /**
     * Retrieves the forgot password link element
     *
     * @return WebElement representing the forgot password link
     */
    public WebElement getForgotPasswordLink() {
        return driver.findElement(forgotPasswordLink);
    }
}
```